

Engr 691/Csci 492: Deep Learning -
Project 3
Part 1: Basic Attention
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Task 1: Hyperparameter Tuning

BLEU Score Evaluation

The model's performance was evaluated using the BLEU score on the same test data as in Part 3 of Project 2. BLEU scores were computed for each configuration to measure translation quality.

Top-Performing Hyperparameter Configurations

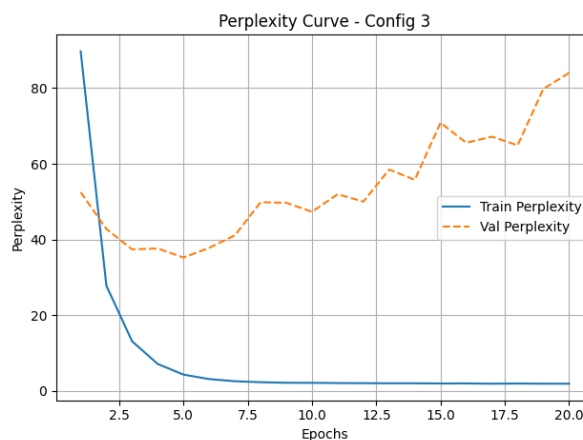
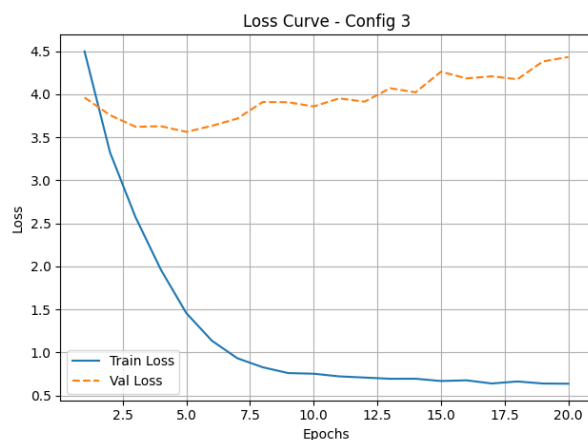
Four configurations were tested using a Bahdanau attention mechanism within an encoder-decoder LSTM architecture. The table below summarizes the top-performing hyperparameter settings:

Config	Embedding Dim	Hidden Size	Learning Rate	Dropout	BLEU Score
C	256	512	0.0005	0.2	0.2552
B	128	256	0.001	0.1	0.2283
A	64	128	0.005	0.1	0.2102

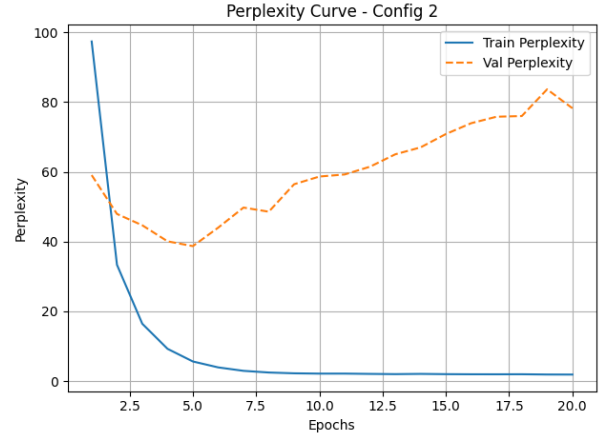
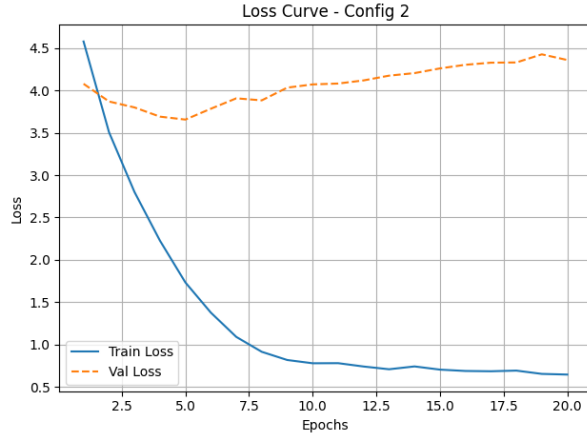
Training and Validation Loss Plots

Each configuration included training and validation loss plots:

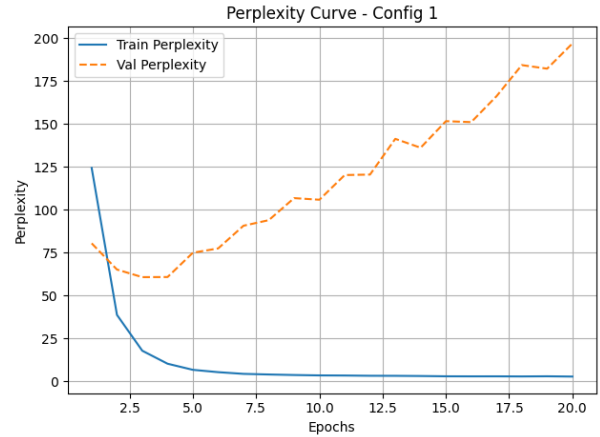
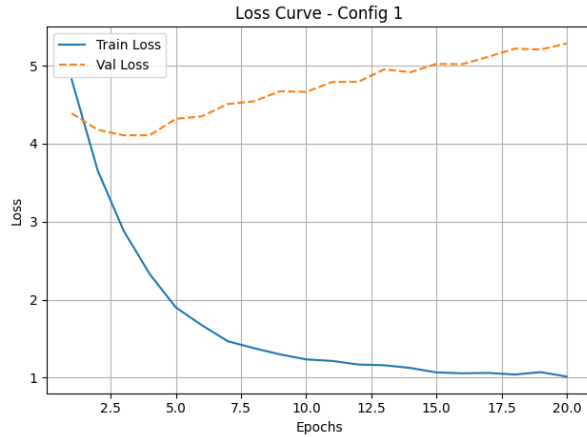
- **Configuration C** (emb_dim=256, hid_dim=512, lr=0.0005, dropout=0.2): Best performing model with high BLEU and a well-behaved validation curve initially.



- **Configuration B** (emb_dim=128, hid_dim=256, lr=0.001, dropout=0.1): Showed stable and low training loss with a gradual rise in validation loss.



- **Configuration A** (emb_dim=64, hid_dim=128, lr=0.005, dropout=0.1): Showed increasing validation loss after epoch 10 indicating early overfitting.



Observation

Effects of different hyperparameter settings based on training results and BLEU scores for the top-performing configurations A, B, and C:

- **Hidden Size:** Increasing the hidden size significantly improved performance. Configuration C, with a hidden size of 512, achieved the highest BLEU score (0.2552). This indicates that a larger model capacity helped capture richer representations for translation.
- **Learning Rate:** A lower learning rate was especially important for larger models. Configuration C used a smaller learning rate (0.0005), which contributed to stable convergence. In contrast, Configuration A had a higher learning rate (0.005) and experienced early overfitting.
- **Dropout:** A dropout of 0.1 worked well for Configurations A and B, while a slightly higher dropout (0.2) in Configuration C provided a good regularization balance. Very high dropout

(0.3 in Configuration D) led to underperformance, showing that excessive regularization can hurt deeper models.

- **Trade-offs:** Configuration A converged quickly but overfit, while Configuration B offered better generalization. Configuration C struck the best balance between expressiveness and generalization, making it the top-performing setup.
- **Overall:** The best results were achieved with moderate dropout, lower learning rates, and increased hidden size. Overfitting was more likely with high learning rates or smaller models.

Outputs:

Config 1: {'emb_dim': 64, 'hid_dim': 128, 'n_layers': 1, 'dropout': 0.1, 'lr': 0.005}

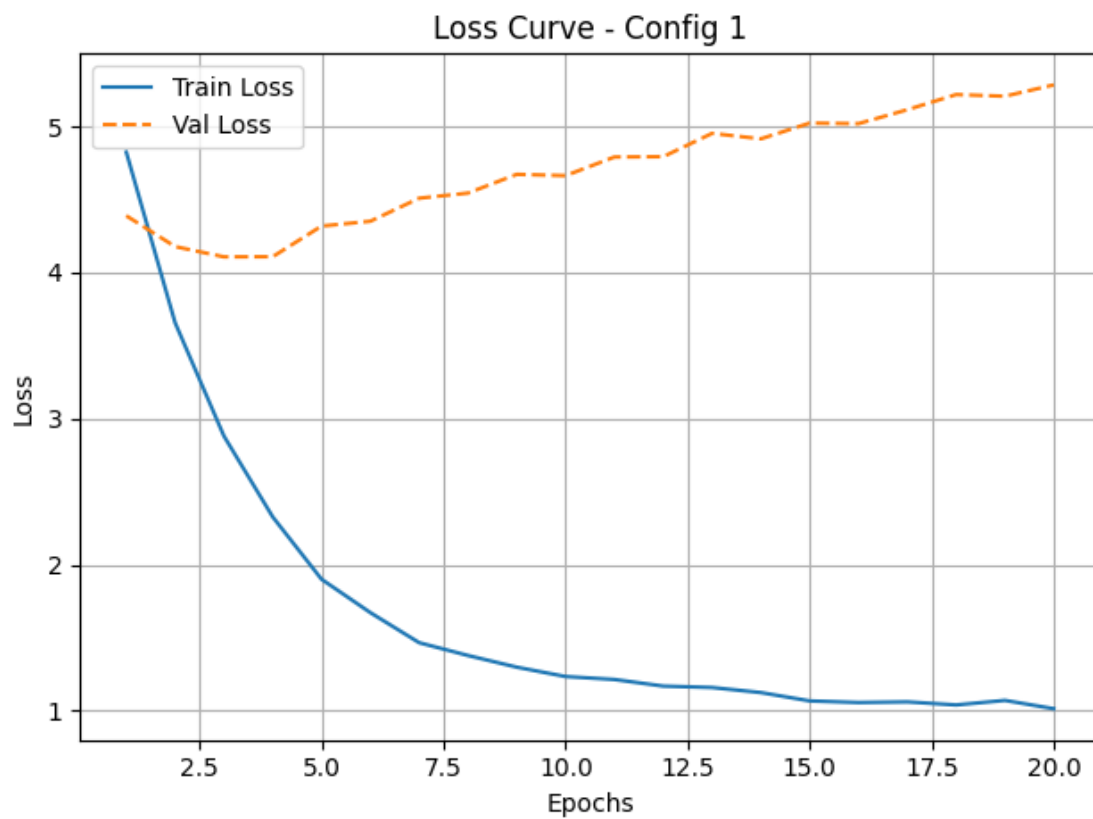
Training model for Config 1...

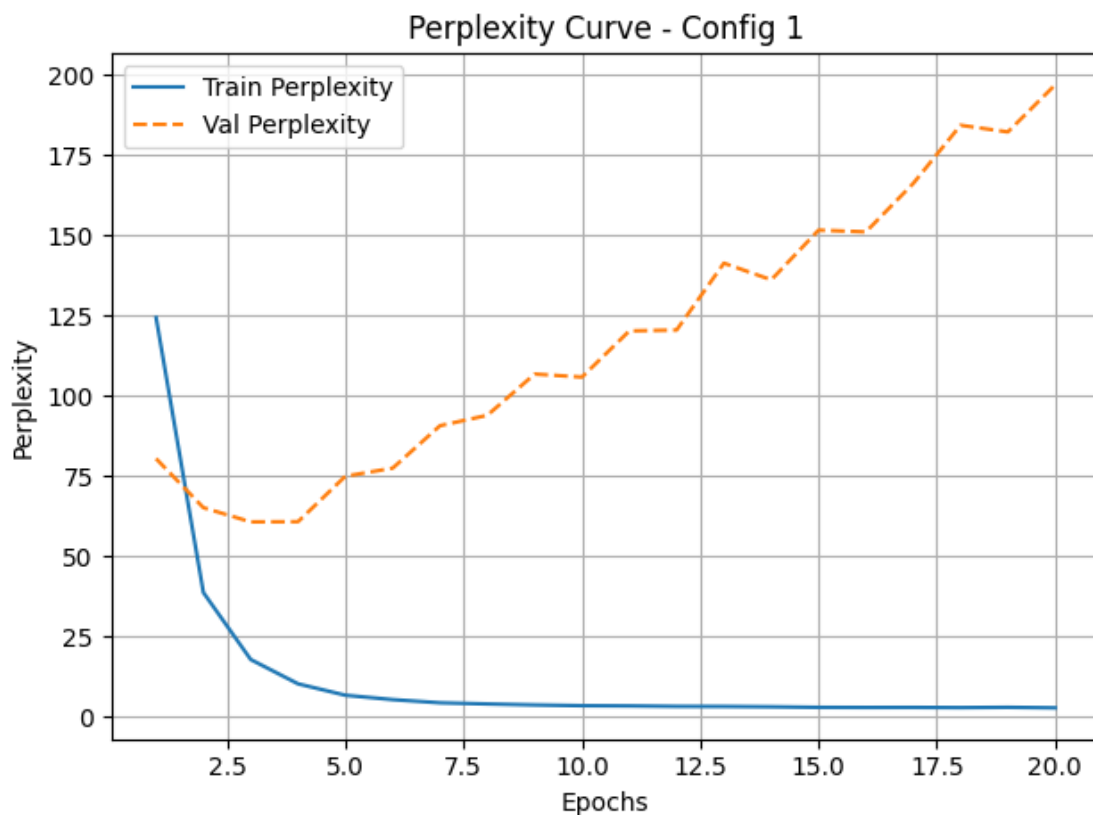
```
Epoch 1 - Train Loss: 4.8239 | Perplexity: 124.44
Epoch 1 - Val Loss: 4.3879 | Val Perplexity: 80.47
Epoch 2 - Train Loss: 3.6563 | Perplexity: 38.72
Epoch 2 - Val Loss: 4.1771 | Val Perplexity: 65.17
Epoch 3 - Train Loss: 2.8807 | Perplexity: 17.83
Epoch 3 - Val Loss: 4.1067 | Val Perplexity: 60.75
Epoch 4 - Train Loss: 2.3270 | Perplexity: 10.25
Epoch 4 - Val Loss: 4.1078 | Val Perplexity: 60.81
Epoch 5 - Train Loss: 1.9005 | Perplexity: 6.69
Epoch 5 - Val Loss: 4.3173 | Val Perplexity: 74.98
Epoch 6 - Train Loss: 1.6720 | Perplexity: 5.32
Epoch 6 - Val Loss: 4.3504 | Val Perplexity: 77.51
Epoch 7 - Train Loss: 1.4663 | Perplexity: 4.33
Epoch 7 - Val Loss: 4.5080 | Val Perplexity: 90.74
Epoch 8 - Train Loss: 1.3786 | Perplexity: 3.97
Epoch 8 - Val Loss: 4.5431 | Val Perplexity: 93.98
Epoch 9 - Train Loss: 1.2987 | Perplexity: 3.66
Epoch 9 - Val Loss: 4.6713 | Val Perplexity: 106.84
Epoch 10 - Train Loss: 1.2346 | Perplexity: 3.44
Epoch 10 - Val Loss: 4.6624 | Val Perplexity: 105.89
Epoch 11 - Train Loss: 1.2145 | Perplexity: 3.37
Epoch 11 - Val Loss: 4.7899 | Val Perplexity: 120.29
Epoch 12 - Train Loss: 1.1693 | Perplexity: 3.22
Epoch 12 - Val Loss: 4.7923 | Val Perplexity: 120.58
Epoch 13 - Train Loss: 1.1596 | Perplexity: 3.19
Epoch 13 - Val Loss: 4.9517 | Val Perplexity: 141.41
Epoch 14 - Train Loss: 1.1250 | Perplexity: 3.08
Epoch 14 - Val Loss: 4.9143 | Val Perplexity: 136.23
Epoch 15 - Train Loss: 1.0678 | Perplexity: 2.91
Epoch 15 - Val Loss: 5.0220 | Val Perplexity: 151.72
Epoch 16 - Train Loss: 1.0569 | Perplexity: 2.88
Epoch 16 - Val Loss: 5.0184 | Val Perplexity: 151.17
```

Epoch 17 - Train Loss: 1.0613 | Perplexity: 2.89
Epoch 17 - Val Loss: 5.1136 | Val Perplexity: 166.27
Epoch 18 - Train Loss: 1.0404 | Perplexity: 2.83
Epoch 18 - Val Loss: 5.2173 | Val Perplexity: 184.43
Epoch 19 - Train Loss: 1.0713 | Perplexity: 2.92
Epoch 19 - Val Loss: 5.2059 | Val Perplexity: 182.34
Epoch 20 - Train Loss: 1.0146 | Perplexity: 2.76
Epoch 20 - Val Loss: 5.2834 | Val Perplexity: 197.04

Evaluating BLEU score...

Config 1 BLEU Score: 0.2102 | Time taken: 0.04 seconds





Source: go.
Predicted: allons-y !
Target: va !

Source: hi.
Predicted: salut !
Target: salut !

Source: run!
Predicted: cours !
Target: cours !

Source: run!
Predicted: cours !
Target: courez !

Source: who?
Predicted: qui ?
Target: qui ?

Source: wow!

Predicted: ça alors !
Target: ça alors !

Source: fire!
Predicted: au joue !
Target: au feu !

Source: help!
Predicted: à l'aide !
Target: à l'aide !

Source: jump.
Predicted: saute.
Target: saute.

Source: stop!
Predicted: stop !
Target: ça suffit !

Config 2: {'emb_dim': 128, 'hid_dim': 256, 'n_layers': 1, 'dropout': 0.1, 'lr': 0.001}

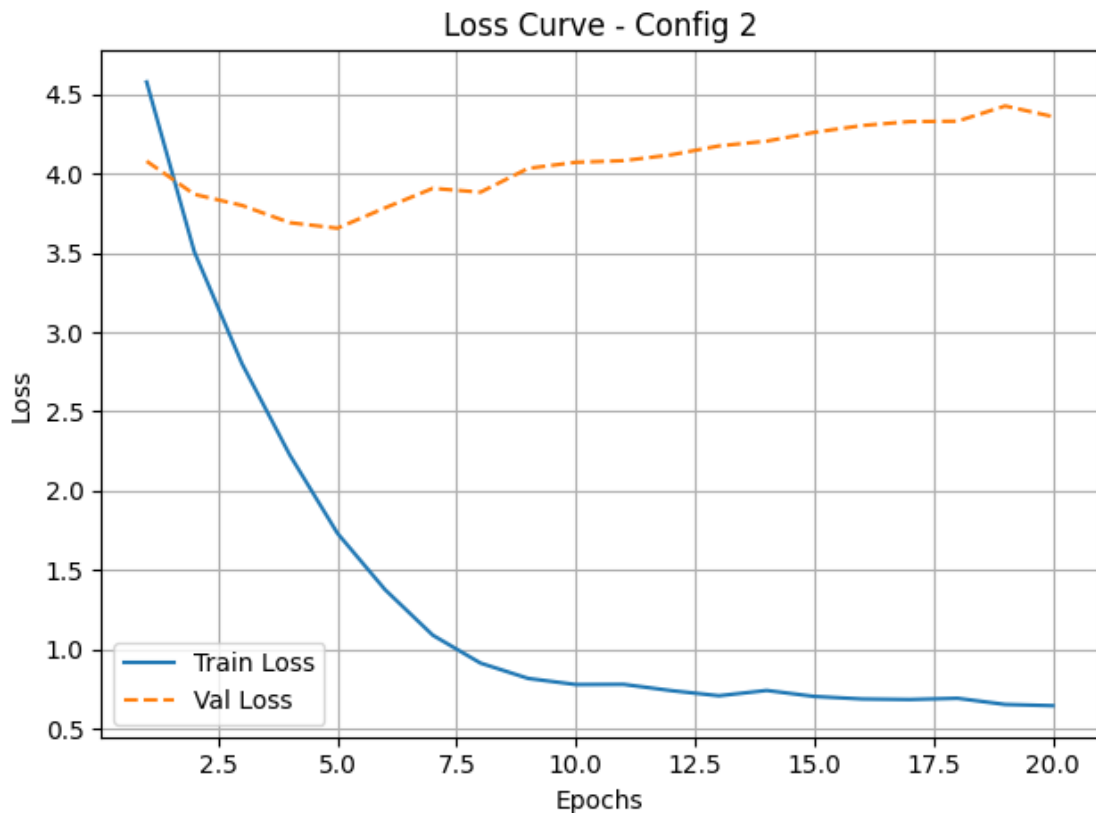
Training model for Config 2...

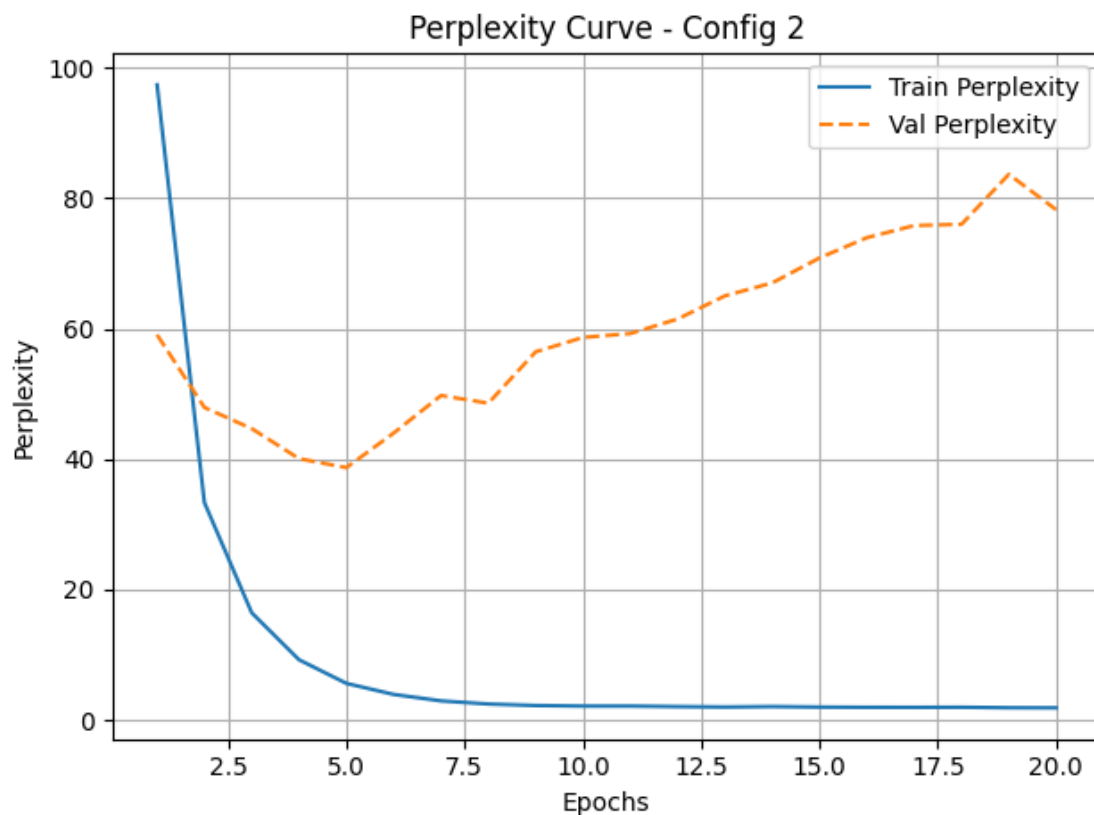
Epoch 1	- Train Loss: 4.5782	Perplexity: 97.34
Epoch 1	- Val Loss: 4.0784	Val Perplexity: 59.05
Epoch 2	- Train Loss: 3.5071	Perplexity: 33.35
Epoch 2	- Val Loss: 3.8695	Val Perplexity: 47.92
Epoch 3	- Train Loss: 2.8014	Perplexity: 16.47
Epoch 3	- Val Loss: 3.7994	Val Perplexity: 44.68
Epoch 4	- Train Loss: 2.2270	Perplexity: 9.27
Epoch 4	- Val Loss: 3.6908	Val Perplexity: 40.08
Epoch 5	- Train Loss: 1.7328	Perplexity: 5.66
Epoch 5	- Val Loss: 3.6559	Val Perplexity: 38.70
Epoch 6	- Train Loss: 1.3768	Perplexity: 3.96
Epoch 6	- Val Loss: 3.7841	Val Perplexity: 43.99
Epoch 7	- Train Loss: 1.0898	Perplexity: 2.97
Epoch 7	- Val Loss: 3.9070	Val Perplexity: 49.75
Epoch 8	- Train Loss: 0.9139	Perplexity: 2.49
Epoch 8	- Val Loss: 3.8829	Val Perplexity: 48.57
Epoch 9	- Train Loss: 0.8172	Perplexity: 2.26
Epoch 9	- Val Loss: 4.0331	Val Perplexity: 56.44
Epoch 10	- Train Loss: 0.7787	Perplexity: 2.18
Epoch 10	- Val Loss: 4.0714	Val Perplexity: 58.64
Epoch 11	- Train Loss: 0.7798	Perplexity: 2.18
Epoch 11	- Val Loss: 4.0816	Val Perplexity: 59.24
Epoch 12	- Train Loss: 0.7403	Perplexity: 2.10
Epoch 12	- Val Loss: 4.1185	Val Perplexity: 61.46

Epoch 13 - Train Loss: 0.7080 | Perplexity: 2.03
Epoch 13 - Val Loss: 4.1746 | Val Perplexity: 65.01
Epoch 14 - Train Loss: 0.7414 | Perplexity: 2.10
Epoch 14 - Val Loss: 4.2047 | Val Perplexity: 67.00
Epoch 15 - Train Loss: 0.7036 | Perplexity: 2.02
Epoch 15 - Val Loss: 4.2602 | Val Perplexity: 70.82
Epoch 16 - Train Loss: 0.6877 | Perplexity: 1.99
Epoch 16 - Val Loss: 4.3031 | Val Perplexity: 73.93
Epoch 17 - Train Loss: 0.6843 | Perplexity: 1.98
Epoch 17 - Val Loss: 4.3278 | Val Perplexity: 75.78
Epoch 18 - Train Loss: 0.6920 | Perplexity: 2.00
Epoch 18 - Val Loss: 4.3304 | Val Perplexity: 75.98
Epoch 19 - Train Loss: 0.6532 | Perplexity: 1.92
Epoch 19 - Val Loss: 4.4264 | Val Perplexity: 83.63
Epoch 20 - Train Loss: 0.6453 | Perplexity: 1.91
Epoch 20 - Val Loss: 4.3593 | Val Perplexity: 78.20

Evaluating BLEU score...

Config 2 BLEU Score: 0.2283 | Time taken: 0.01 seconds





Source: go.
Predicted: va !
Target: va !

Source: hi.
Predicted: salut !
Target: salut !

Source: run!
Predicted: cours !
Target: cours !

Source: run!
Predicted: cours !
Target: courez !

Source: who?
Predicted: qui ?
Target: qui ?

Source: wow!

Predicted: ça alors !
Target: ça alors !

Source: fire!
Predicted: au feu !
Target: au feu !

Source: help!
Predicted: à l'aide l'aide !
Target: à l'aide !

Source: jump.
Predicted: saute.
Target: saute.

Source: stop!
Predicted: stop !
Target: ça suffit !

Config 3: {'emb_dim': 256, 'hid_dim': 512, 'n_layers': 1, 'dropout': 0.2, 'lr': 0.0005}

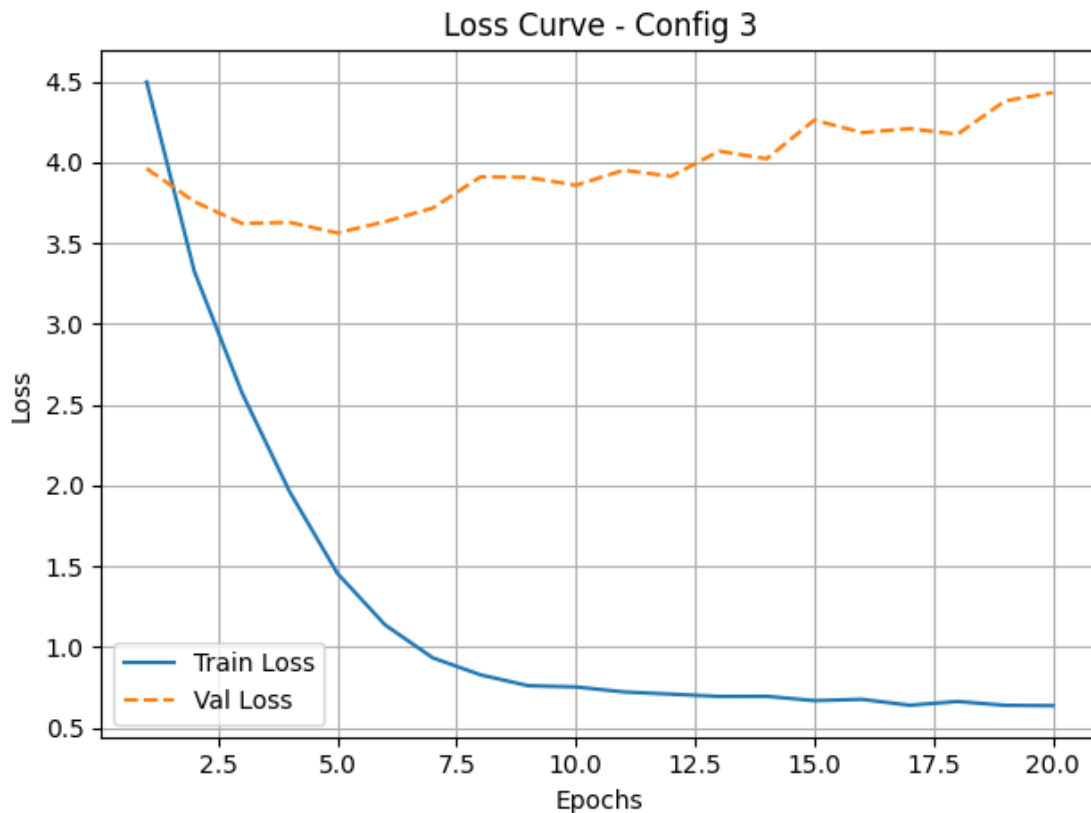
Training model for Config 3...

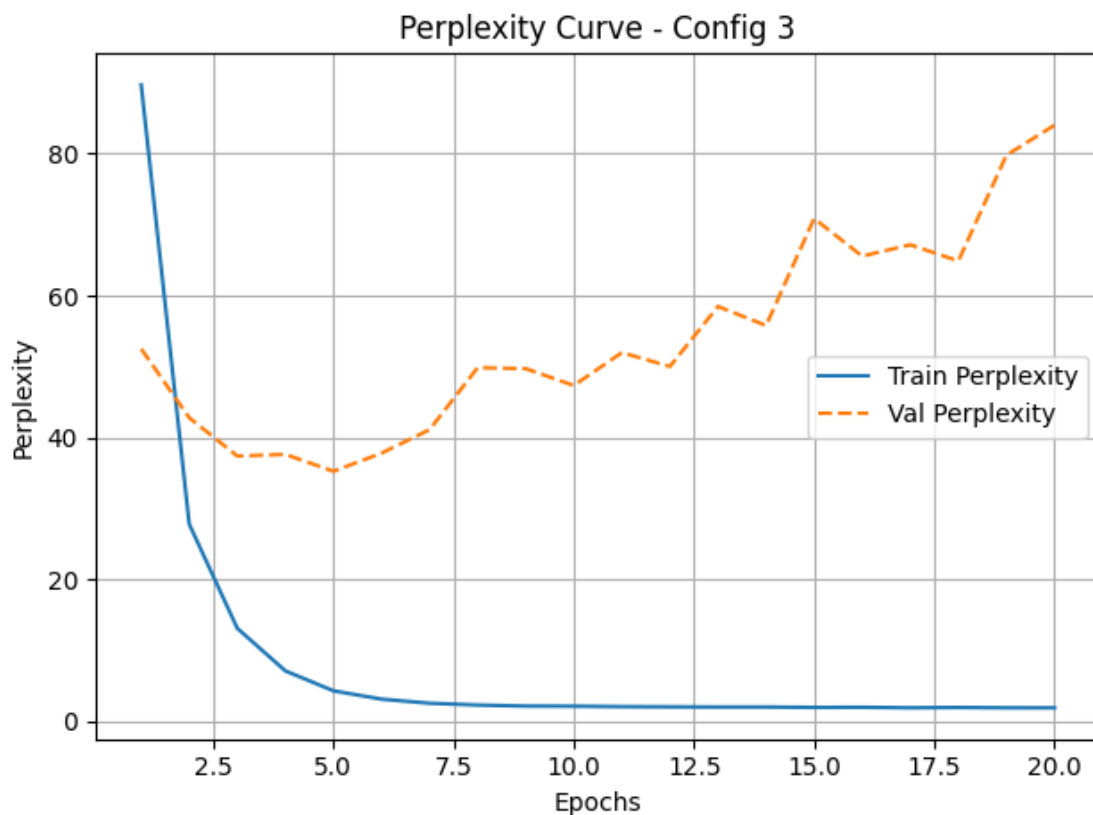
Epoch 1 - Train Loss:	4.4967		Perplexity:	89.72
Epoch 1 - Val Loss:	3.9608		Val Perplexity:	52.50
Epoch 2 - Train Loss:	3.3245		Perplexity:	27.78
Epoch 2 - Val Loss:	3.7563		Val Perplexity:	42.79
Epoch 3 - Train Loss:	2.5728		Perplexity:	13.10
Epoch 3 - Val Loss:	3.6214		Val Perplexity:	37.39
Epoch 4 - Train Loss:	1.9604		Perplexity:	7.10
Epoch 4 - Val Loss:	3.6276		Val Perplexity:	37.62
Epoch 5 - Train Loss:	1.4564		Perplexity:	4.29
Epoch 5 - Val Loss:	3.5623		Val Perplexity:	35.24
Epoch 6 - Train Loss:	1.1375		Perplexity:	3.12
Epoch 6 - Val Loss:	3.6322		Val Perplexity:	37.80
Epoch 7 - Train Loss:	0.9335		Perplexity:	2.54
Epoch 7 - Val Loss:	3.7162		Val Perplexity:	41.11
Epoch 8 - Train Loss:	0.8286		Perplexity:	2.29
Epoch 8 - Val Loss:	3.9090		Val Perplexity:	49.85
Epoch 9 - Train Loss:	0.7611		Perplexity:	2.14
Epoch 9 - Val Loss:	3.9061		Val Perplexity:	49.71
Epoch 10 - Train Loss:	0.7537		Perplexity:	2.12
Epoch 10 - Val Loss:	3.8573		Val Perplexity:	47.34
Epoch 11 - Train Loss:	0.7228		Perplexity:	2.06
Epoch 11 - Val Loss:	3.9507		Val Perplexity:	51.97
Epoch 12 - Train Loss:	0.7091		Perplexity:	2.03
Epoch 12 - Val Loss:	3.9124		Val Perplexity:	50.02

Epoch 13 - Train Loss: 0.6950 | Perplexity: 2.00
Epoch 13 - Val Loss: 4.0687 | Val Perplexity: 58.48
Epoch 14 - Train Loss: 0.6956 | Perplexity: 2.00
Epoch 14 - Val Loss: 4.0215 | Val Perplexity: 55.78
Epoch 15 - Train Loss: 0.6690 | Perplexity: 1.95
Epoch 15 - Val Loss: 4.2608 | Val Perplexity: 70.87
Epoch 16 - Train Loss: 0.6770 | Perplexity: 1.97
Epoch 16 - Val Loss: 4.1831 | Val Perplexity: 65.57
Epoch 17 - Train Loss: 0.6404 | Perplexity: 1.90
Epoch 17 - Val Loss: 4.2073 | Val Perplexity: 67.18
Epoch 18 - Train Loss: 0.6637 | Perplexity: 1.94
Epoch 18 - Val Loss: 4.1727 | Val Perplexity: 64.89
Epoch 19 - Train Loss: 0.6399 | Perplexity: 1.90
Epoch 19 - Val Loss: 4.3795 | Val Perplexity: 79.80
Epoch 20 - Train Loss: 0.6376 | Perplexity: 1.89
Epoch 20 - Val Loss: 4.4311 | Val Perplexity: 84.02

Evaluating BLEU score...

Config 3 BLEU Score: 0.2552 | Time taken: 0.01 seconds





Source: go.
Predicted: va !
Target: va !

Source: hi.
Predicted: salut !
Target: salut !

Source: run!
Predicted: cours !
Target: cours !

Source: run!
Predicted: cours !
Target: courez !

Source: who?
Predicted: qui ?
Target: qui ?

Source: wow!

Predicted: ça alors !
Target: ça alors !

Source: fire!
Predicted: au feu !
Target: au feu !

Source: help!
Predicted: à l'aide !
Target: à l'aide !

Source: jump.
Predicted: saute.
Target: saute.

Source: stop!
Predicted: ça !
Target: ça suffit !

Config 4: {'emb_dim': 512, 'hid_dim': 1024, 'n_layers': 2, 'dropout': 0.3, 'lr': 0.0001}

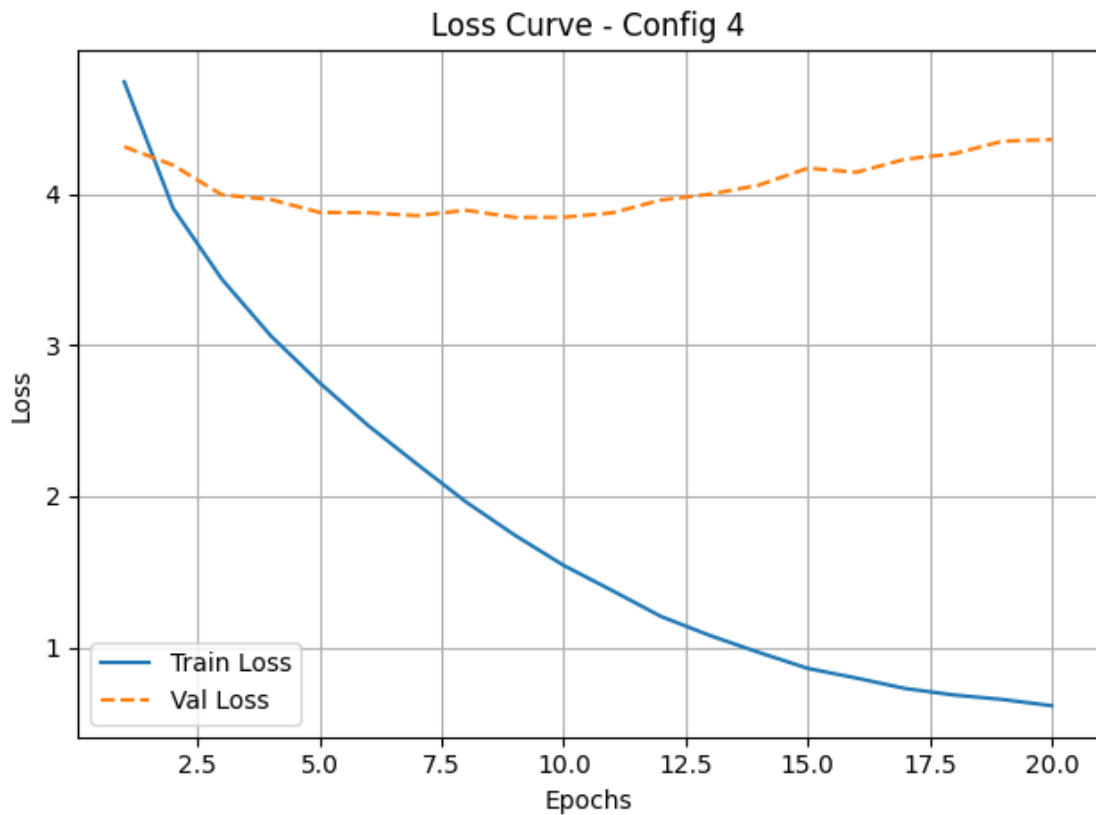
Training model for Config 4...

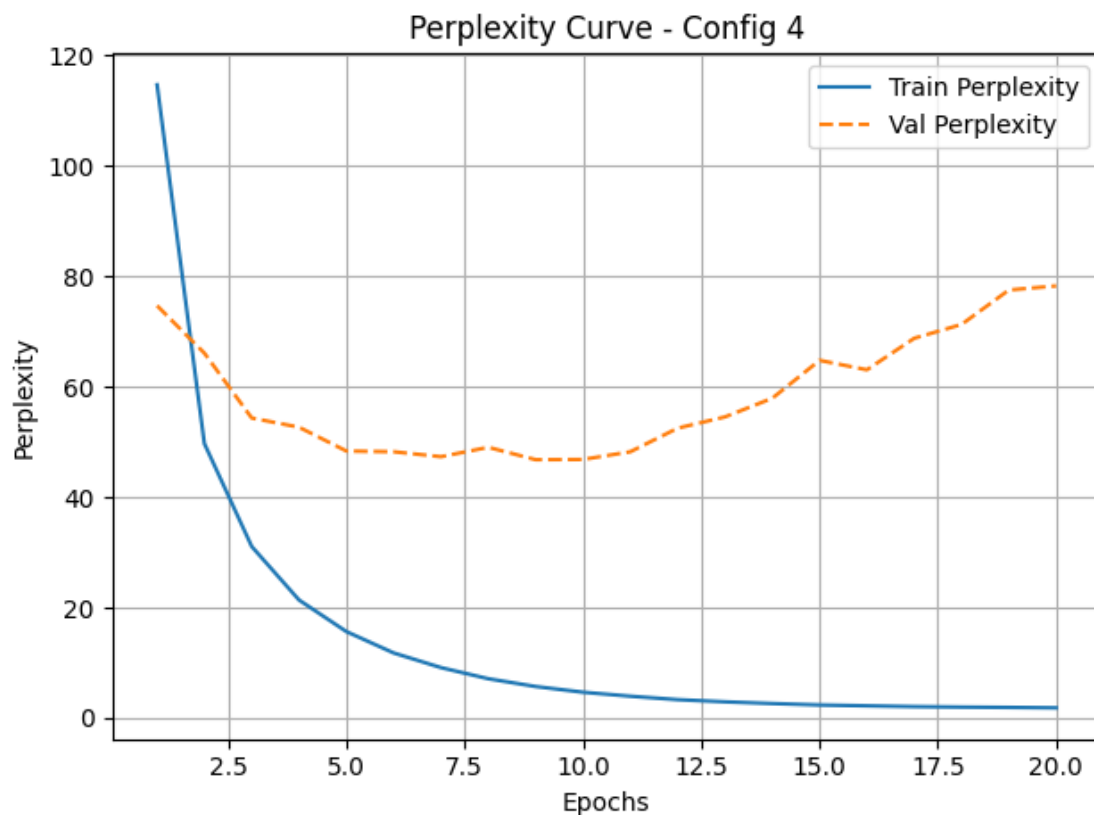
Epoch 1 - Train Loss:	4.7417		Perplexity:	114.63
Epoch 1 - Val Loss:	4.3128		Val Perplexity:	74.65
Epoch 2 - Train Loss:	3.9056		Perplexity:	49.68
Epoch 2 - Val Loss:	4.1904		Val Perplexity:	66.05
Epoch 3 - Train Loss:	3.4357		Perplexity:	31.05
Epoch 3 - Val Loss:	3.9948		Val Perplexity:	54.32
Epoch 4 - Train Loss:	3.0619		Perplexity:	21.37
Epoch 4 - Val Loss:	3.9636		Val Perplexity:	52.65
Epoch 5 - Train Loss:	2.7508		Perplexity:	15.65
Epoch 5 - Val Loss:	3.8782		Val Perplexity:	48.34
Epoch 6 - Train Loss:	2.4666		Perplexity:	11.78
Epoch 6 - Val Loss:	3.8754		Val Perplexity:	48.20
Epoch 7 - Train Loss:	2.2117		Perplexity:	9.13
Epoch 7 - Val Loss:	3.8567		Val Perplexity:	47.31
Epoch 8 - Train Loss:	1.9621		Perplexity:	7.11
Epoch 8 - Val Loss:	3.8921		Val Perplexity:	49.01
Epoch 9 - Train Loss:	1.7420		Perplexity:	5.71
Epoch 9 - Val Loss:	3.8452		Val Perplexity:	46.77
Epoch 10 - Train Loss:	1.5421		Perplexity:	4.67
Epoch 10 - Val Loss:	3.8460		Val Perplexity:	46.81
Epoch 11 - Train Loss:	1.3746		Perplexity:	3.95
Epoch 11 - Val Loss:	3.8754		Val Perplexity:	48.20
Epoch 12 - Train Loss:	1.2032		Perplexity:	3.33
Epoch 12 - Val Loss:	3.9603		Val Perplexity:	52.47

Epoch 13 - Train Loss: 1.0777 | Perplexity: 2.94
Epoch 13 - Val Loss: 3.9984 | Val Perplexity: 54.51
Epoch 14 - Train Loss: 0.9658 | Perplexity: 2.63
Epoch 14 - Val Loss: 4.0585 | Val Perplexity: 57.89
Epoch 15 - Train Loss: 0.8602 | Perplexity: 2.36
Epoch 15 - Val Loss: 4.1706 | Val Perplexity: 64.75
Epoch 16 - Train Loss: 0.7962 | Perplexity: 2.22
Epoch 16 - Val Loss: 4.1438 | Val Perplexity: 63.04
Epoch 17 - Train Loss: 0.7269 | Perplexity: 2.07
Epoch 17 - Val Loss: 4.2305 | Val Perplexity: 68.75
Epoch 18 - Train Loss: 0.6840 | Perplexity: 1.98
Epoch 18 - Val Loss: 4.2662 | Val Perplexity: 71.25
Epoch 19 - Train Loss: 0.6542 | Perplexity: 1.92
Epoch 19 - Val Loss: 4.3501 | Val Perplexity: 77.48
Epoch 20 - Train Loss: 0.6133 | Perplexity: 1.85
Epoch 20 - Val Loss: 4.3596 | Val Perplexity: 78.23

Evaluating BLEU score...

Config 4 BLEU Score: 0.1346 | Time taken: 0.03 seconds





Source: go.
Predicted: va !
Target: va !

Source: hi.
Predicted: salut !
Target: salut !

Source: run!
Predicted: pas !
Target: cours !

Source: run!
Predicted: pas !
Target: courez !

Source: who?
Predicted: qui ?
Target: qui ?

Source: wow!

Predicted: couillon couillon armé. armé. armé. armé. armé. armé. armé. couillon
 armé. armé. couillon armé. couillon armé. couillon armé. couillon couillon
 Target: ça alors !

Source: fire!
 Predicted: au feu !
 Target: au feu !

Source: help!
 Predicted: est fou. fou. vivant. vivant. vivant. vivant. vivant. vivant. vivant.
 vivant. vivant. vivant. vivant. vivant. vivant. vivant. vivant. vivant.
 Target: à l'aide !

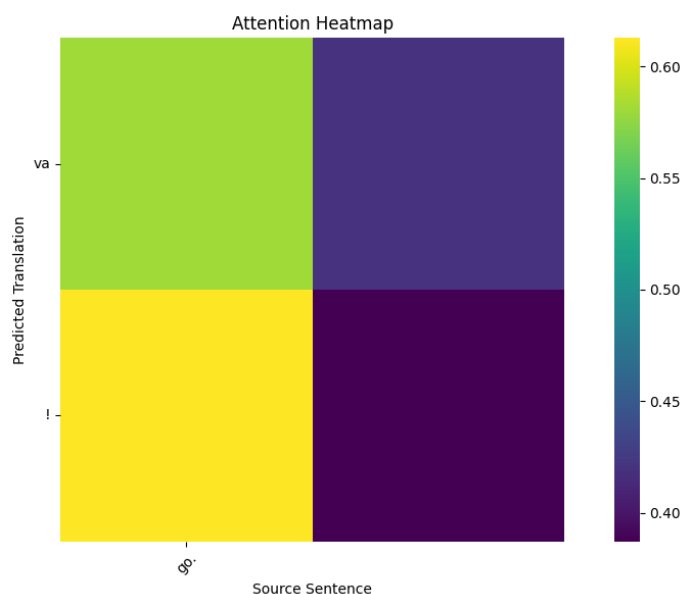
Source: jump.
 Predicted: le là.
 Target: saute.

Source: stop!
 Predicted: pas sûr.
 Target: ça suffit !

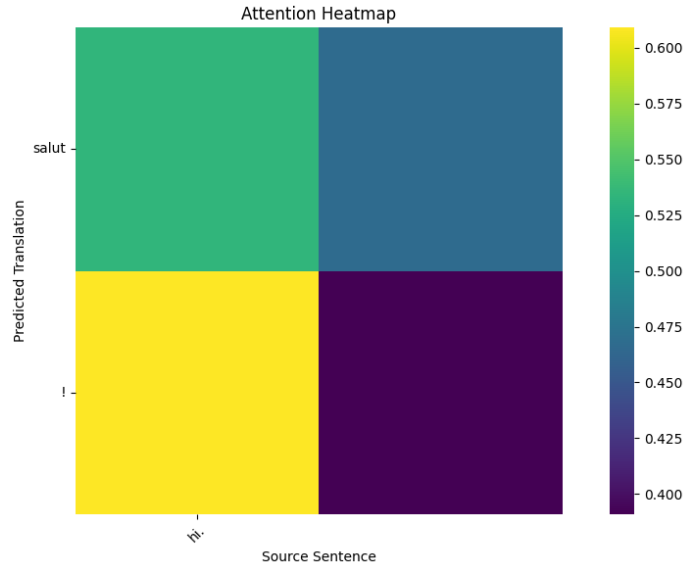
Task 2:

The attention heatmaps reveal how the model aligns source and predicted tokens for each sentence. These plots offer insight into the behavior of the attention mechanism.

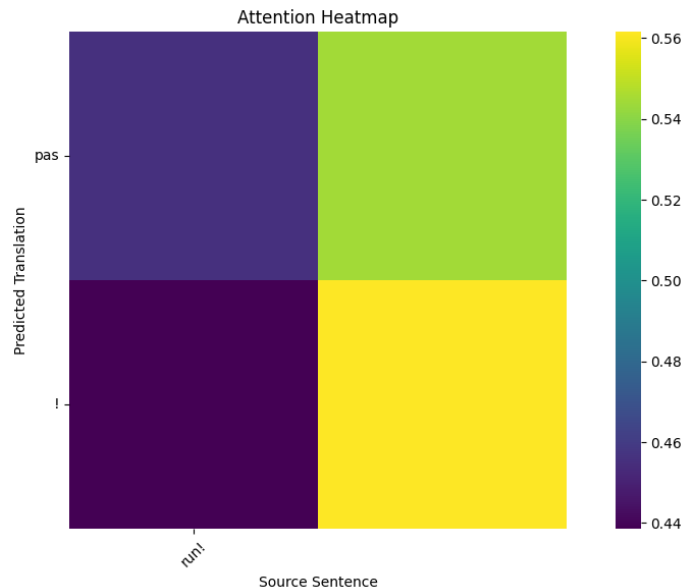
- **Example 1 (go. → va !):** The attention is sharply focused on the token go, correctly aligning it with va. The second token ! is also aligned appropriately. This indicates that the model has learned a precise mapping for short imperative sentences.



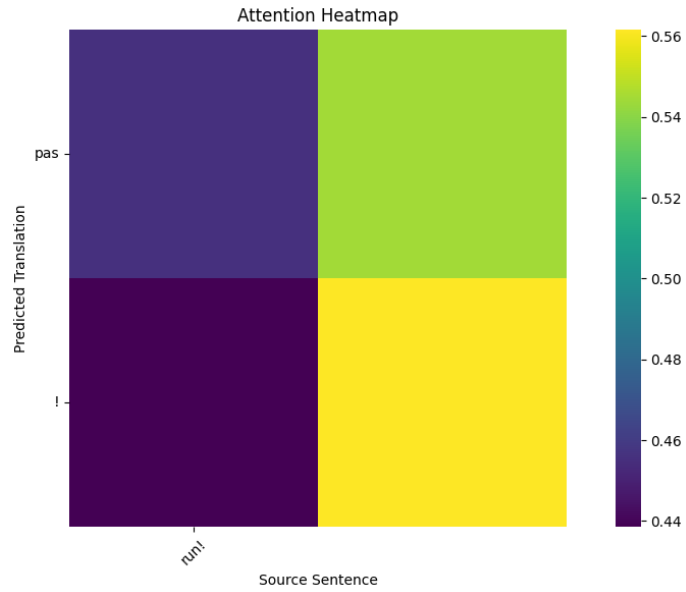
- **Example 2 (hi. → salut !):** The attention weights are strong and clearly focused on the word `hi`, guiding the decoder to output `salut`. The alignment is strong and direct, reflecting confident predictions.



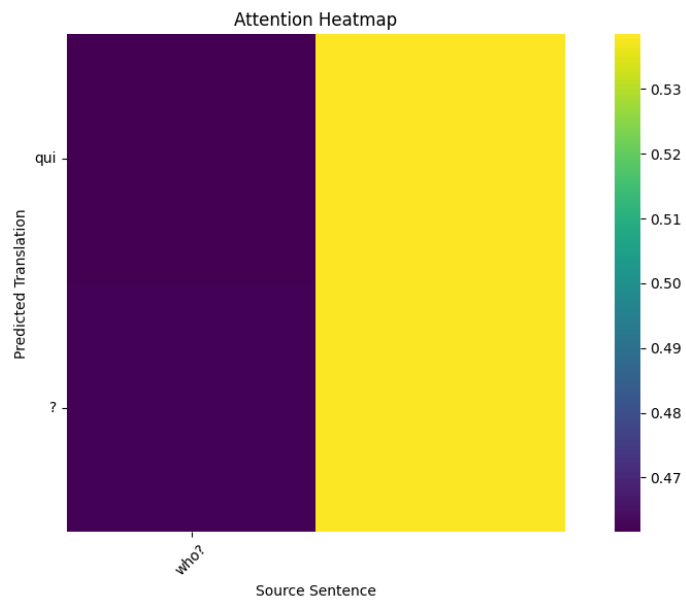
- **Example 3 (run! → pas !):** This is a failure case. The correct translation would have been `cours !` or `courez !`, but the model outputs `pas !`, which is semantically incorrect. The attention weights appear unfocused, suggesting confusion in aligning the imperative verb.



- **Example 4 (run! → pas !):** Similar to Example 3, the model again outputs `pas !` incorrectly. Though the attention is consistent, it likely reflects memorization of incorrect pairs from training data rather than true translation. The source-to-target alignment is present but fails semantically.



- **Example 5 (who? → qui ?):** This is a successful translation. The attention is sharply focused on the token **who**, which maps correctly to **qui**. The model handles the question form well and maintains correct punctuation alignment too.



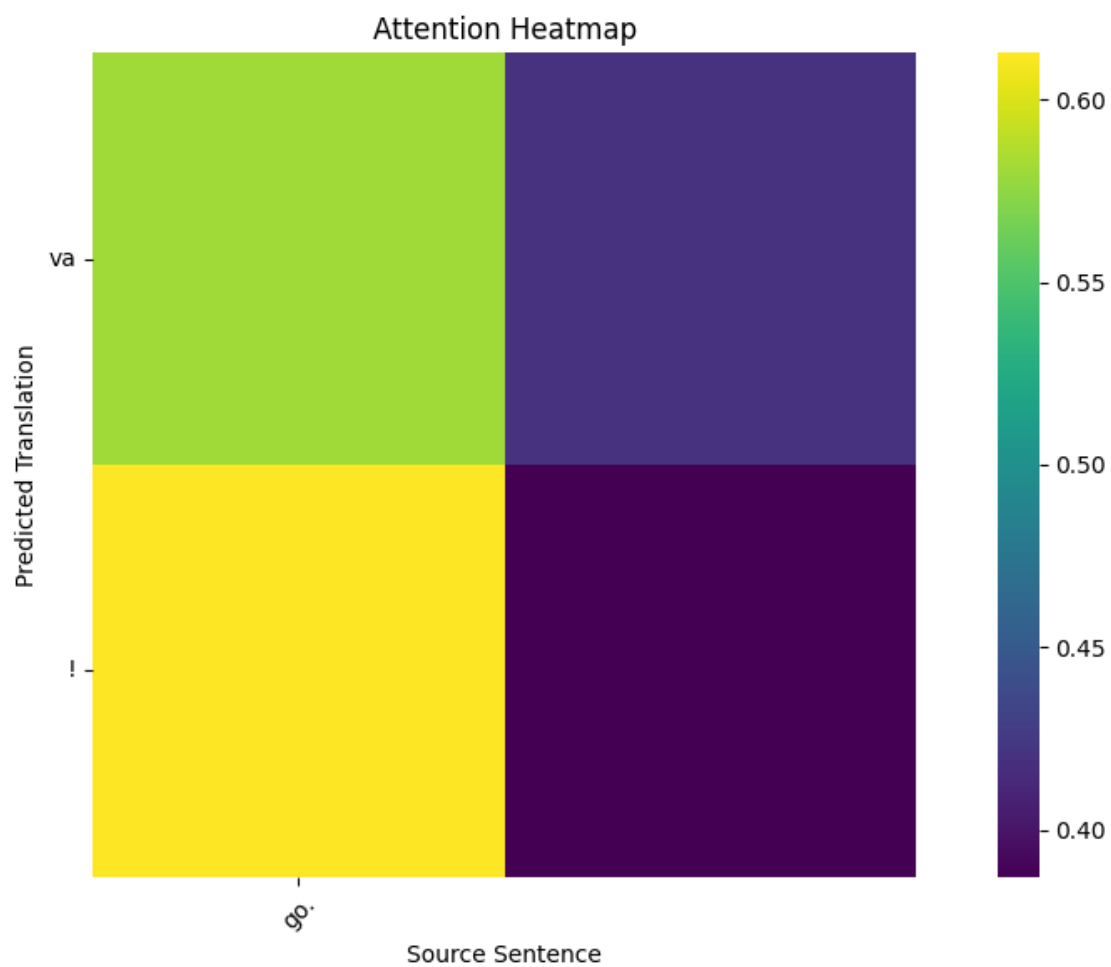
Heatmap Outputs:

Example 1

Source: go.

Predicted: va !

Target: va !

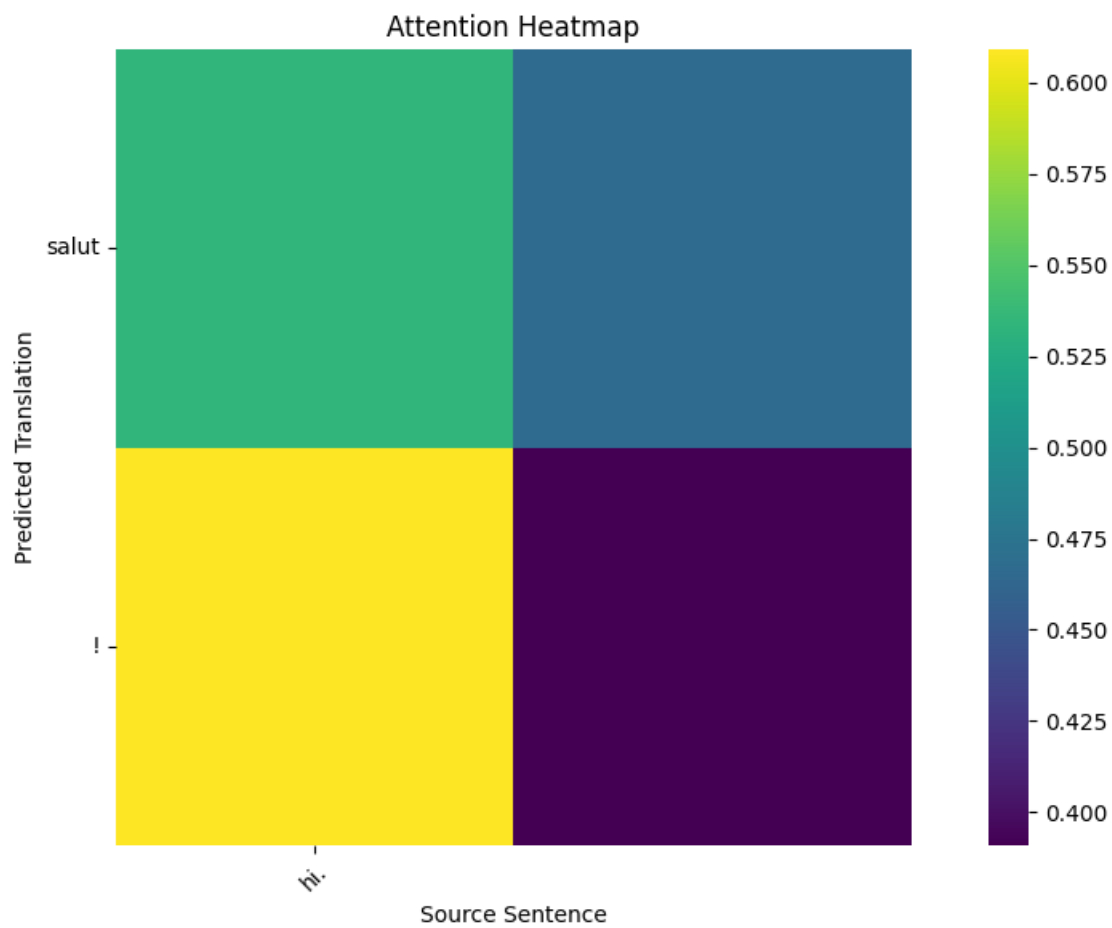


Example 2

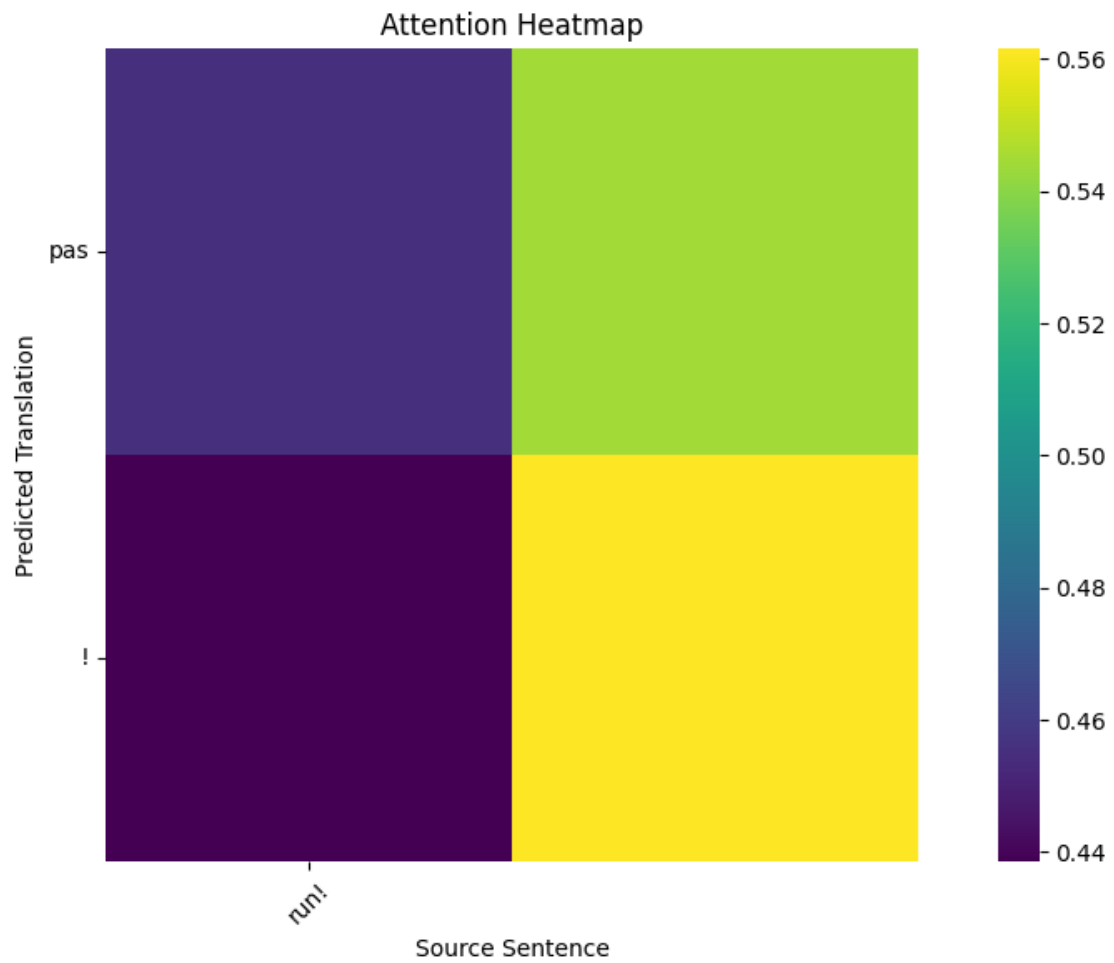
Source: hi.

Predicted: salut !

Target: salut !



Example 3
Source: run!
Predicted: pas !
Target: cours !

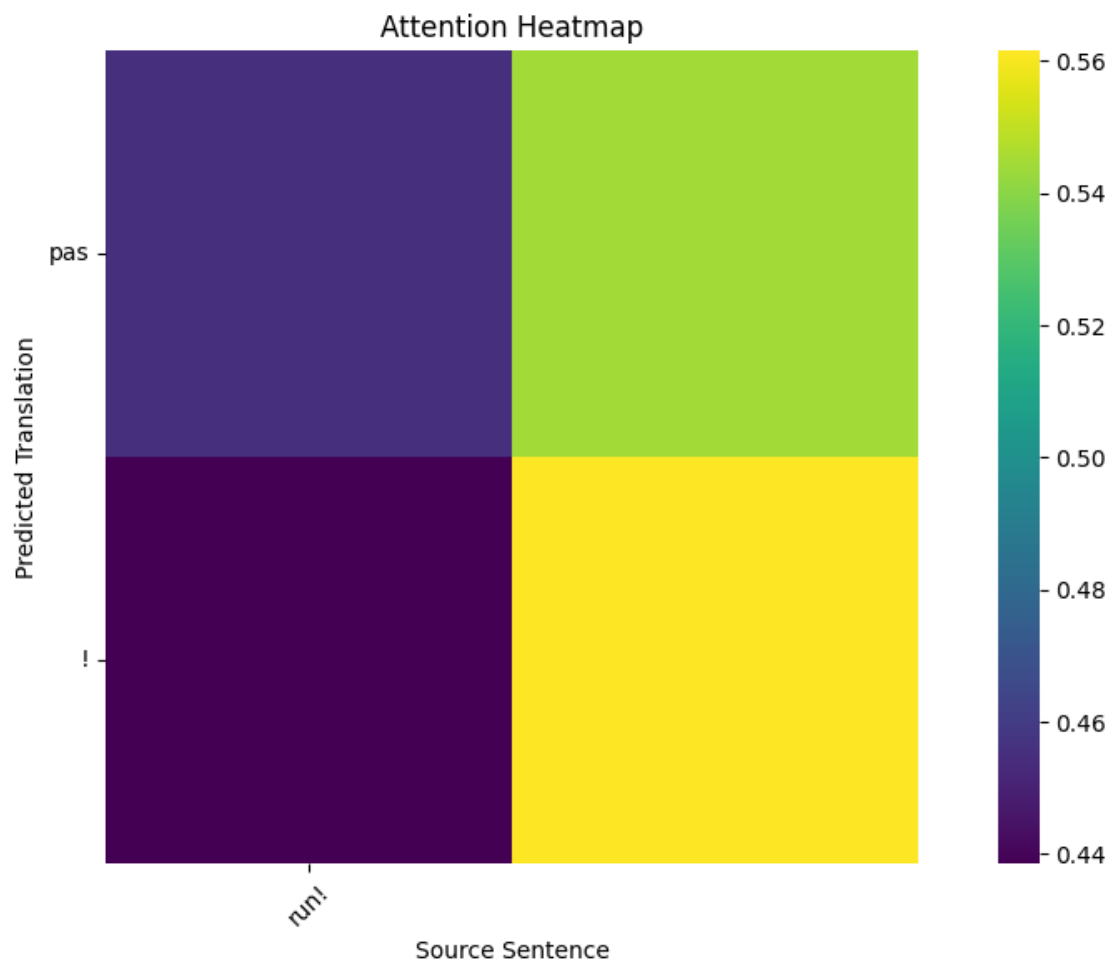


Example 4

Source: run!

Predicted: pas !

Target: courez !



Example 5

Source: who?

Predicted: qui ?

Target: qui ?

